Masked Spectral–Spatial Feature Prediction for Hyperspectral Image Classification

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***Abstract*— Transformer has emerged as a preferred method for hyperspectral (HS) image classification due to its ability to model long-range dependency. Whereas the transformer contains numerous parameters and further available labeled HS data is limited, which makes it difficult to get a well-trained transformer. Accordingly, we propose a novel HS image classification method called masked spectral–spatial feature prediction (MSSFP). It aims at helping the transformer understand the complicated spectral–spatial structures without labeled HS data, further improving the classification performance. Specifically, the input HS cube is first divided into two sequences along spectral and spatial dimensions, respectively. Then, a portion of these two sequences are masked out and we train a transformer-based encoder–decoder network to predict the hand-crafted features of masked regions. After pretraining, the encoder is fine-tuned to derive two classification results from input spectral and spatial sequences. Finally, spectral and spatial results are aggregated adaptively based on uncertainty comparison. In comparison experiments, MSSFP outperforms several state-of-the-art HS image classification methods on three benchmark datasets including Indian Pines (IP), Houston (HU), and Pavia University (PUS).**

***Index Terms*— Hyperspectral (HS) image classification, masked feature prediction, transformer.**

I. Introduction

HYPERSPECTRAL (HS) sensors could capture hundreds of bands covering almost the whole electromagnetic spectrum for specific land-cover objects. The generated HS image provides abundant spectral information for land-cover objects, making it possible to identify these objects. HS image classification task is meant to assign a predefined semantic label to each HS pixel. Due to this task, HS image is widely applied to various fields, such as mineral exploration, precision agriculture, and environmental monitoring. At first, the HS vector of each pixel is directly used to infer the corresponding label by traditional classifiers, such as support vector machine (SVM) [1] and k-nearest neighbor (k-NN) [2]. Later, several works proposed to extract hand-crafted features (e.g., morphological profiles [3]) from HS images for classification. However, these hand-crafted features limit the classification performance because they are too "shallow" to represent spectral–spatial information of HS images [4].

Deep learning has been successfully applied in the remote sensing community like HS image classification [5], [6], [7], [8], [9], [10], [11], [12], [13]. The first attempt was made by Chen et al. [14] to propose a multilayer autoencoder for HS image classification via spectral information reconstruction. Similarly, a 1-D convolutional neural network (CNN) was proposed to extract spectral features [15]. These methods significantly improve the classification performance because they can capture the nonlinear structure of HS data. Besides, to model the correlation between any two spectral bands, Mou et al. [16] regarded the HS vector of each pixel as sequential data and used recurrent neural network (RNN) to handle this HS sequence. Hang et al. [17] constructed a cascaded RNN model to explicitly learn redundant and complementary information between spectral bands. However, land-cover objects exhibit high intraclass and low interclass spectral variabilities, and thus these spectral feature-based methods have difficulty in producing satisfactory results.

To enhance the discrimination of spectral features, spatial information is introduced by researches. For example, a stacked autoencoder (SAE) took the spatial patches from the principal component of the HS image as inputs to learn spatial features [14]. Unfortunately, the fully-connected layer in these models demands flattening the input patch into a vector, failing to capture the spatial structure of HS images. Hence, 2-D CNN was used to extract spatial features directly from the original 2-D spatial patch [18]. Based on 2-D CNN, Haut et al. [19] additionally adopted random occlusion to generate more training samples for mitigating the overfitting problem. Most of these methods perform dimensionality reduction on the spectrum before extracting features, leading to the loss of spectral information. Accordingly, researchers proposed to extract features from the original HS image

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without any information compression. The simplest way is to feed the HS cube into 3-D CNN to adequately learn spectral–spatial features [15]. To deal with the decreasing-accuracy phenomenon, Zhong et al. [20] integrated residual connections to 3-D CNN. Besides, several works designed a hybrid network composed of 2-D and 3-D convolutional layers to reduce the complexity of 3-D CNN [21], [22].

It has been shown that not all information provided by the HS image matters for the classification task [23], which demands the model to highlight the related information and suppress the irrelated one. To this end, Haut et al. [24] used a visual attention mechanism to recognize the most representative regions in HS images. In [25], spectral and spatial attention are conducted consecutively to reveal the significance of all bands and all neighbor pixels. To learn the long-range dependency between nonadjacent spectral bands, Mei et al. [26] integrated spectral–spatial attention mechanism into bidirectional long short-term memory (LSTM). Roy et al. [27] used an attention mechanism to adaptively determine the optimal receptive field of 3-D convolutional kernels for capturing both fine- and coarse-grained structures. These methods perform spectral and spatial attention on the intermediate feature successively while some other methods design a two-branch network to complete all attention processes simultaneously. For example, a two-branch spectral–spatial attention network was designed to extract detailed features for cross-domain HS image classification [28]. Different from the above methods, Yu et al. [29] inferred attention maps using the semantic information from higher layers in a feedback way. However, attention-based methods are incapable of learning the correlation between HS pixels within the HS image, which hinders them from obtaining better performance.

Self-attention was first proposed to learn global dependencies between local features in the field of natural language processing [30]. Recently, self-attention has been successfully used to model the relationships between any two pixels within the image for various computer vision tasks [31]. Similarly, for each target pixel in the HS image, Sun et al. [32] attempted to introduce self-attention to eliminate the negative effect of irrelevant pixels within the input patch. The result demonstrates that self-attention works for HS image processing. Zhou et al. [33] constructed a self-mutual attention module to model the correlation between spectral and spatial domains. To determinate the optimal structure of the spectral–spatial self-attention network, a factorized architecture search was proposed in [34]. Since the extended version (transformer [30]) of self-attention has been successfully applied to natural image processing [35], researches attempted to directly use the transformer to classify HS images. Sun et al. [36] used a convolutional module to extract spectral–spatial features which were subsequently fed into the transformer to obtain semantic features. In [37], convolutional operations were integrated into the transformer to simultaneously extract spectral–spatial features. Peng et al. [38] designed a cross-attention module that was plugged into the transformer to perform feature interaction between spectral and spatial domains.

Generally, there still exist convolutional layers for feature extraction in the above methods, yet the local connection of convolutional layers impedes the capturing of long-range dependency. Thus, several works simply used transformers to build classification models without any convolutional operations. In [39], for each HS pixel, the spatial patch extracted from each band was flattened into a vector and then successively fed into the transformer along the spectral dimension. Hong et al. [40] proposed group-wise spectral embedding which regarded a group of bands rather than individual bands as a token to learn locally detailed information. It is well-known that the transformer contains numerous parameters, which demand a lot of training samples to update them. Nevertheless, labeled HS images are limited in practice due to the high acquisition cost. Consequently, transformer-based HS image classification methods would inevitably encounter the generalization problem (e.g., overfitting). In this article, masked spectral–spatial feature prediction (MSSFP) is proposed to pretrain a spectral–spatial transformer via *mask-and-predict*, and then we fine-tune it with labeled HS images. At the pretraining stage, a portion of two sequences extracted from spectral and spatial dimensions are randomly masked, and the transformer is trained to predict the hand-crafted features [e.g., principal components and local binary pattern (LBP)] of masked regions. With respect to the fine-tuning stage, we feed input spectral and spatial sequences into the transformer to infer two classification results, and an adaptive aggregation strategy is adopted to fuse them based on uncertainty comparison.

The contributions of our work are summarized as follows.

1) We propose an HS image classification method called MSSFP. MSSFP obtains a well-trained transformer by predicting hand-crafted features of masked spectral and spatial regions in HS images. Although labeled training samples are limited, this transformer achieves competitive classification performance after fine-tuning.
2) To consider the correlation between spectral and spatial domains, we fuse two classification results separately derived from spectral and spatial features by using an adaptive aggregation strategy. This strategy compares uncertainties of spectral and spatial results to reveal the correlation which subsequently guides how to fuse these two results.
3) The proposed MSSFP is compared with several state-of-the-art classification methods on three public datasets including Indian Pines (IP), Houston (HU), and Pavia University (PUS). The experimental results demonstrate that MSSFP achieves the best performance with the limited training samples.

II. Related Work

A. Self-Supervised HS Image Classification

Deep-learning methods have dramatically boosted the performance of HS image classification, in comparison with traditional methods. These methods are based on supervised learning which demands a huge number of training samples. However, it is difficult to acquire labeled HS images in

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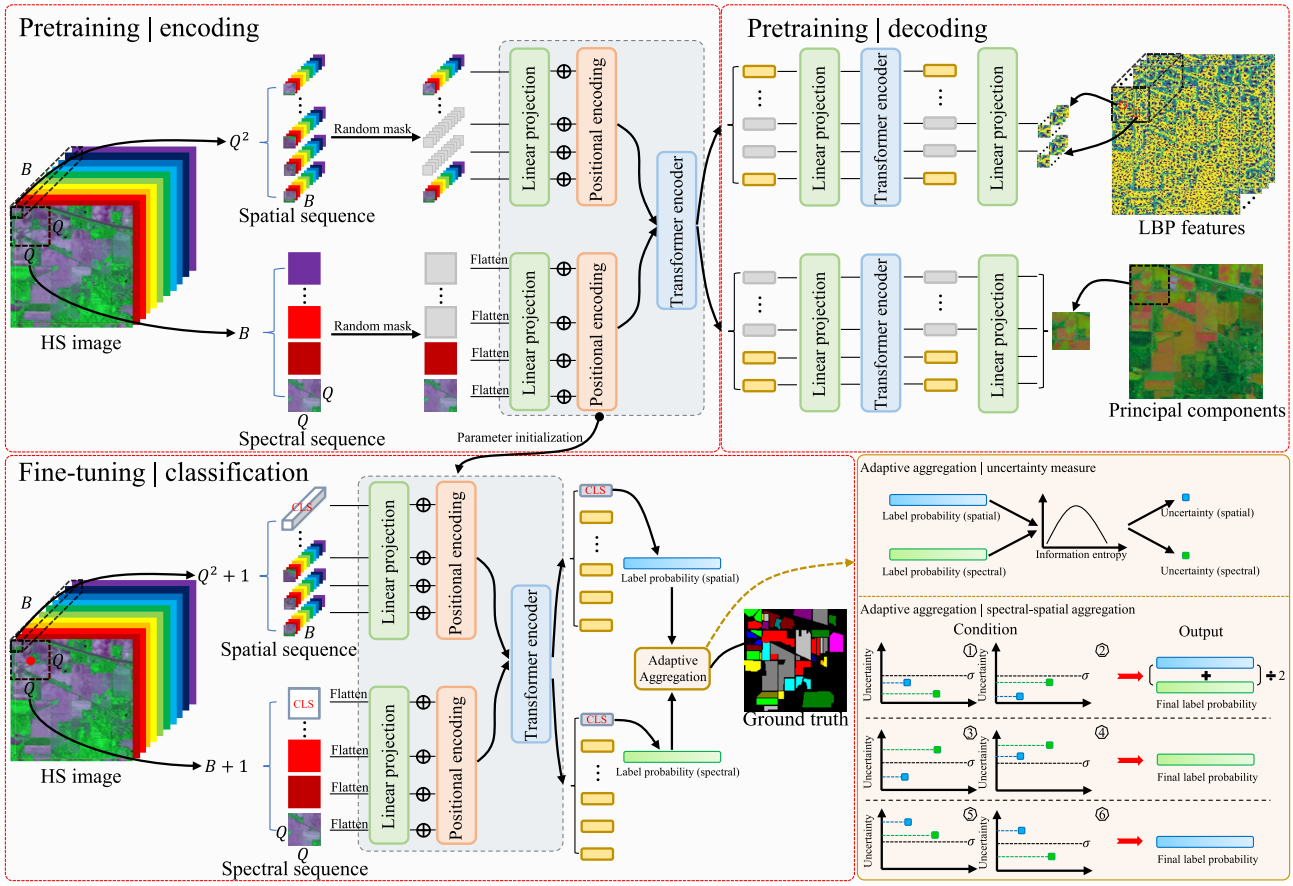

Fig. 1. Flowchart of proposed MSSFP.

reality. Therefore, self-supervised learning is proposed to learn features from unlabeled HS images, aiming at improving the classification performance. According to [41], self-supervised learning methods for HS image classification could be divided into two classes: generative methods and contrastive methods. Generative methods apply various transformations to input data for generating pseudo labels which could be used to train deep-learning methods. For instance, Yue et al. [42] rotated each HS cube by specific angles and then a CNN was trained to predict the rotation angle of each input. Similarly, the performance of HS image classification benefitted from the image rotation prediction task in [43]. Yang et al. [44] flipped the spectral order of each input HS cube and performed geometric transformations with different scales on it. Then, they trained a CNN to predict the spectral order and scale class of each generated HS cube.

Different from generative methods which focus on the intraimage relationship, contrastive methods pay more attention to the interimage relationship. The goal of contrastive methods is to minimize similarities between different views of the same sample and maximize those between separate views of different samples. In [45], stochastic data augmentations were performed on multiscale patches centered at a target pixel to generate positive samples, and augmented views of multiscale patches centered at other pixels were regarded as negative samples. InfoNCE [46] loss function was then used to pull positive samples close to each other and push negative samples away from each other. Similarly, several works [47], [48] followed this strategy to fully explore spectral and spatial information in a self-supervised manner. Nevertheless, the above methods might mistake same-category samples for negative sample pairs, easily affecting feature extraction. To handle this problem, pseudo labels were proposed to be used to identify true-negative samples [49], [50]. Concretely, unsupervised methods like k-means were used to cluster samples for obtaining pseudo labels. Then, samples with the same pseudo labels were treated as positive sample pairs and vice versa.

B. Mask-and-Predict Task

Mask-and-predict task is a popular self-supervised learning method that has emerged recently. It was proposed for natural language processing and indeed helps language models understand the context of sentences better. More specifically, given a sentence, *mask-and-predict* method first randomly masks some words and then compels the transformer to predict masked words by unmasked ones. *Mask-and-predict* task has attracted attention from the research in the field of image processing lately. He et al. [51] made a successful attempt to apply *mask-and-predict* task for natural image processing via the proposed MAE. They split the original image into several small patches and 75% of the patches were masked out. The remaining patches were successively fed into the

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transformer to reconstruct the masked patches. Based on MAE, MAEST [52] was proposed to reconstruct masked spectral and spatial regions of HS images for learning generalizing noise-free representations. To learn instance discrimination information, Huang et al. [4] proposed SC-SS-MTr combining contrastive learning and *mask-and-predict* task. They treated masked and unmasked views of the same HS cube as positive sample pairs and masked or unmasked views of different HS cubes were regarded as negative sample pairs. Both MAEST and SC-SS-MTr reconstruct the original pixels of masked regions which might lead to the ambiguous problem [53]. Different from them, we propose a spectral–spatial transformer to predict hand-crafted features of masked spectral and spatial regions in HS images.

III. Masked Spectral–Spatial Feature Prediction

The proposed MSSFP is illustrated in Fig. 1. There exist two stages in MSSFP: pretraining and fine-tuning. During the pretraining stage, input spectral and spatial sequences are extracted from the HS vector and spatial patch, respectively. Then, they are fed into a transformer-based encoder to obtain corresponding features and random masking is performed on both spectral and spatial feature sequences. These two masked feature sequences are separately fed into two transformer-based decoders to produce the hand-crafted features of masked regions. When the pretraining is done, we use the same input spectral and spatial sequences to infer two classification results via the above well-trained encoder. Finally, adaptive aggregation is adopted to fuse spectral and spatial results based on uncertainty comparison.

A. Pretraining

Most of the existing transformer-based methods for HS image classification could not perform at their best, because it is hard to train the parameters randomly initialized well with the limited training samples. If the parameters of the transformer could be pretrained instead of random initialization, the difficulty of training might be reduced. Motivated by a pretraining task named *mask-and-predict* [51] for natural images, it occurs to us that this task could be used to get a well-trained transformer using unlabeled HS images. Significantly, different from recent *mask-and-predict*-based methods MAEST [52] and SC-SS-MTr [4], we propose to predict the hand-crafted features rather than pixel values of masked regions for alleviating the ambiguous problem. Besides, to exploit the correlation between spectral and spatial domains, MSSFP derives two results from spectral and spatial domains, respectively, and they are further fused via uncertainty comparison.

Let the cube centered at the target pixel x be $\mathbf{X} \in \mathbb{R}^{Q\times Q\times B}$ where $Q \times Q$ is the spatial size and B is the number of bands. The spectral sequence $\mathbf{S_c} = \{s_c^i | s_c^i \in \mathbb{R}^{Q\times Q}, 1 \le i \le B\}$ and spatial sequence $\mathbf{S_a} = \{s_a^i | s_a^i \in \mathbb{R}^B, 1 \le i \le Q^2\}$ are extracted from the cube $\mathbf{X}$ along spectral and spatial dimensions, respectively. Taking the spatial sequence as an example, we illustrate how to predict the hand-crafted features of masked regions. First of all, linear projection is performed on each element s_a^i to generate the embedding sequence

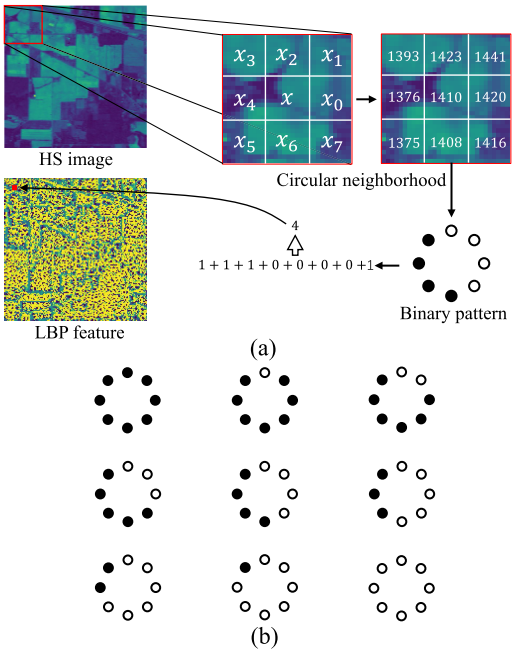

Fig. 2. Illustration of (a) LBP features of HS images and (b) uniform binary patterns.

$$\mathbf{E_a} = \left\{ e_a^i | e_a^i \in \mathbb{R}^{C_e} = s_a^i \mathbf{W_e} + p_a^i, 1 \le i \le Q^2 \right\} \tag{1}$$

where $\mathbf{W_e} \in \mathbb{R}^{B\times C_e}$ represents the weight matrix and the positional vector $p_a^i \in \mathbb{R}^{C_e}$ is used for positional encoding. C_e denotes the dimension of the encoding feature. p_a^i is calculated by

$$p_a^i(j) = \begin{cases} \sin\left(\dfrac{i}{10\,000^{\frac{2j}{C_e}}}\right), & j = 2n-1 \\ \cos\left(\dfrac{i}{10\,000^{\frac{2j}{C_e}}}\right), & j = 2n, \end{cases} \quad n = 1, 2, \ldots, \frac{C_e}{2}. \tag{2}$$

Next, we generate a random mask sequence $\mathbf{M_a} = \{m_a^i | m_a^i \in \{0, 1\}, 1 \le i \le Q^2\}$ where 0 indicates mask and 1 indicates no mask. Similar to [51], the masking ratio is set to 75%. Based on $\mathbf{M_a}$, the ith element in $\mathbf{E_a}$ will be eliminated when the corresponding value m_a^i is 0 and vice versa.

Given the masked sequence $\mathbf{E'_a}$, we feed it into a transformer-based encoder to get the feature sequence containing long-range dependencies. Specifically, each layer in the transformer consists of layer normalization (LN) [54], multihead self-attention (MSA), and multilayer perceptron (MLP). The output $\mathbf{O^k}$ of kth layer is formulated as follows:

$$\begin{aligned} \hat{\mathbf{O}}^{\mathbf{k}} &= \mathrm{MSA}\left(\mathrm{LN}\left(\mathbf{O^{k-1}}\right)\right) + \mathbf{O^{k-1}} \\ \mathbf{O^k} &= \mathrm{MLP}\left(\mathrm{LN}\left(\hat{\mathbf{O}}^{\mathbf{k}}\right)\right) + \hat{\mathbf{O}}^{\mathbf{k}}. \end{aligned} \tag{3}$$

Note that, the input sequence $\mathbf{O^0}$ of the first layer equals to $\mathbf{E'_a}$. The output $\mathbf{O_a} = \{o_a^i | o_a^i \in \mathbb{R}^{C_e}, 1 \le i \le M_a\}$, where $M_a = \lfloor 0.75Q^2 \rfloor$, of the last layer of transformer encoder is regarded as the feature sequence of $\mathbf{E'_a}$. Then, the length of $\mathbf{O_a}$ is recovered to B by inserting trainable tokens into the

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masked positions. Similar to (1), $\mathbf{O_a}$ is transformed to $\mathbf{D_a}$ via linear projection and positional encoding

$$\mathbf{D_a} = \left\{ d_a^i \middle| d_a^i \in \mathbb{R}^{C_d} = o_a^i \mathbf{W_d} + q_a^i, 1 \le i \le Q^2 \right\} \tag{4}$$

where $\mathbf{W_d} \in \mathbb{R}^{C_e \times C_d}$ represents the weight matrix and the positional vector $q_a^i \in \mathbb{R}^{C_d}$ is also obtained by (2). C_d denotes the dimension of the decoding feature. Once $\mathbf{D_a}$ is produced, it is fed into a transformer-based decoder to reconstruct the hand-crafted features of masked regions. The structure of the decoder is exactly the same as that of the encoder. According to $\mathbf{M_a}$, the predictions corresponding to the masked region could be extracted from the output sequence of the decoder. By concatenating them together, the prediction $\mathbf{F_a} \in \mathbb{R}^{M_a \times B}$ of masked region is obtained.

Similarly, the prediction $\mathbf{F_c} \in \mathbb{R}^{M_b \times Q^2}$, where $M_b = \lfloor 0.75B \rfloor$, of the masked region in spectral dimension, is inferred by following the above procedure. At this point, $\mathbf{F_a}$ and $\mathbf{F_c}$ are all we can get. However, it is difficult to acquire the ground truth corresponding to $\mathbf{F_a}$ and $\mathbf{F_c}$ for training. A simple solution [4], [52] is to map these two embeddings to the original HS pixels and thus the input cube could be directly regraded as the ground truth. Whereas pixel-wise reconstruction tasks have difficulty in recovering lost high-frequency information (e.g., texture) because the reconstruction loss (e.g., mean squared error) would encourage producing pixel-wise averages of plausible solutions which are overly-smooth [53]. To this end, we take two hand-crafted features (i.e., principal components and LBP) rather than HS pixels as the ground truth for spectral and spatial domains. LBP describes the spatial textures of an image by comparing each pixel with its neighbor pixels. Principal components are usually utilized to represent spectral features and they could be directly acquired by principal component analysis (PCA). Herein, we will elaborate on the generation of LBP features. Generally, the orientations of same-category land-cover objects vary with different scenes. To handle this phenomenon, uniform LBP [55] which has the characteristic of rotation invariance is introduced to measure spatial textures of HS images. Specifically, the target pixel x in the marked region in Fig. 2(a) is taken as an example. Besides, similar to [55], the number P of neighbor pixels and the radius R of the circular neighborhood are empirically set to 8 and 1, respectively. As a result, eight neighbor pixels $\{x_i | 0 \le i \le 7\}$ are selected and each of them is compared with x using a sign function f_s

$$f_s(x_i) = \begin{cases} 1, & x_i \ge x \\ 0, & x_i < x. \end{cases} \tag{5}$$

In the circular binary pattern, black and white circles separately indicate 0 and 1 values which are calculated by f_s. The traditional LBP feature is defined as a decimal number (i.e., 225_{10}) corresponding to a binary sequence (i.e., $11100001_2/(x_0 \rightarrow x_7)$). However, if the image is rotated, the value of x_0 will change and then the LBP feature will be different. To make LBP feature rotation invariant, rotation invariant binary patterns are designed. For each possible binary pattern V, circular neighborhood is rotated clockwise by eight angles $\{(45n)^\circ | 1 \le n \le 8\}$ and eight LBP features are generated. The minimum of these eight LBP features is selected as an index for V. Binary patterns that share the same index are grouped into the same rotation-invariant binary pattern. Consequently, 2^8 possible binary patterns are reduced to 36 unique rotation-invariant binary patterns. In order to further improve the computational efficiency, the spatial transition is used to reduce the number of possible binary patterns. The spatial transition T is defined as

$$T = |f_s(x_7 - x) - f_s(x_0 - x)| + \sum_{i=1}^{7} |f_s(x_i - x) - f_s(x_{i-1} - x)|. \tag{6}$$

Based on T, the LBP feature is redefined by

$$f_s(x_i) = \begin{cases} \sum_{i=0}^{7} f_s(x_i), & T \le 2 \\ 9, & T > 2. \end{cases} \tag{7}$$

As a result, 36 rotation invariant binary patterns are reduced to 9 uniform binary patterns which are shown in Fig. 2(b). Finally, the LBP feature of x changes from 225 to 4.

When we get the principal components $\mathbf{G_c} \in \mathbb{R}^{Q \times Q \times 3}$ and LBP feature $\mathbf{G_a} \in \mathbb{R}^{Q \times Q \times B}$, they are used to produce the ground truth for spectral and spatial branches. For spatial branch, $\mathbf{G_a}$ is reshaped into $Q^2 \times B$ and divided into a sequence $\{g_a^i | g_a^i \in \mathbb{R}^B, 1 \le i \le Q^2\}$. The goal of the spatial branch is to predict the LBP features of masked regions. Therefore, $\mathbf{G_a}$ is masked out via $\mathbf{M_a}$ and concatenated into a 2-D matrix $\mathbf{G'_a} \in \mathbb{R}^{M_a \times B}$ which is the final ground truth. Different from the spatial branch, the spectral branch aims at predicting the principal components of all spectral bands. Thus, reshaped $\mathbf{G_c} \in \mathbb{R}^{Q^2 \times 3}$ is directly set as the ground truth of the spectral branch. The loss function of the pretraining stage is formulated as

$$\text{loss} = \sum \left[\left(\mathbf{F_a W_1} - \mathbf{G'_a} \right)^2 + \left(\mathbf{F'_c W_2} - \mathbf{G_c} \right)^2 \right] \tag{8}$$

where $\mathbf{W_1} \in \mathbb{R}^{B \times B}$ and $\mathbf{W_2} \in \mathbb{R}^{M_b \times 3}$ are weight matrixes, and $\mathbf{F'_c}$ is the transposition of $\mathbf{F_c}$.

B. Fine-Tuning

After the pretraining stage, the transformer-based encoder is able to extract long-range relationships in both spectral and spatial domains from HS images. That is to say, this encoder could be regarded as a well-trained spectral–spatial feature extractor. However, we cannot directly use it to realize HS image classification because it was trained for a *mask-and-predict* task. We just need to fine-tune it using training samples labeled for classification. Similar to the pretraining stage, the spectral sequence $\mathbf{S_c}$ and spatial sequence $\mathbf{S_a}$ are extracted for the target pixel x. Note that, motivated by [35], an extra classification token ("CLS" in Fig. 1) needs to be inserted at the end of the input sequence. After linear projection and positional encoding, the spectral embedding $\mathbf{H_c} = \{h_c^i | h_c^i \in \mathbb{R}^{C_e}, 1 \le i \le Q^2 + 1\}$ and spatial embedding $\mathbf{H_a} = \{h_a^i | h_a^i \in \mathbb{R}^{C_e}, 1 \le i \le B + 1\}$ are obtained. Next, these two embedding

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sequences are fed into the encoder to derive spectral and spatial features via (3). Different from the pretraining stage, we set the number of hidden nodes of the last MLP in the transformer to be the same as the number K of categories. Besides, only the last element (i.e., classification token) of the feature sequence is fed into the last MLP. Consequently, we get the spectral- and spatial-based predictions $j_c \in \mathbb{R}^K$ and $j_a \in \mathbb{R}^K$. Given the real label $j \in \mathbb{R}^K$, the loss function is defined as

$$\text{loss} = -\sum_{i}^{K} \left(j^i \log j_a^i + j^i \log j_c^i \right). \tag{9}$$

Note that, the initial parameters of linear projection, positional encoding, and encoder are inherited from the pretraining stage.

Adaptive Aggregation: As is known to all, same-category land-cover objects might have various spectral signals while they share similar spatial information. In other words, these objects could be identified more correctly only by spatial features. Whereas sometimes, the relationship between spectral and spatial domains is complementary rather than redundant. In this case, if spectral and spatial information are fused, the classification accuracy would be improved. It is therefore significant to exploit the relationship between spectral and spatial domains. However, most of the existing methods directly fuse spectral and spatial predictions by weighted summation [33] without figuring out the correlation between them. Different from these methods, we derive two indicators from spectral- and spatial-based classification results. The indicator could tell us whether the corresponding domain provides useful information or interference information. Consequently, we are able to know whether to use both spectral and spatial results or just one of them for better classification performance.

Specifically, uncertainty is regarded as an indicator, and information entropy is used to measure the uncertainty of classification results. The indicators of spectral and spatial domains are separately denoted as $u_c = \sum_i^K j_c^i \log_2 j_c^i$ and $u_a = \sum_i^K j_a^i \log_2 j_a^i$. The larger the value of an indicator is, the more sure of the classification result the corresponding domain is and vice versa. In addition, if the value of an indicator is smaller than a threshold σ which is manually set, the corresponding classification result is ambiguous and unable to be adopted. As shown in Fig. 1, all six possible situations are listed when classifying HS images. When both u_c and u_a are smaller than σ, the relationship between the two domains is regarded as complementary. Thus, the final result is defined as $(j_c + j_a)/2$ to aggregate spectral and spatial information. If u_a is larger than both σ and u_c or u_c is larger than both σ and u_a, there exists redundant information between the two domains. In this case, the classification result corresponding to a larger indicator value is selected as the final result.

IV. Experiments

In this section, three public datasets including IP, HU, and PUS are first introduced and then we provide the implementation details. Next, we compare MSSFP with several state-of-the-art classification methods on these three datasets to validate the effectiveness of MSSFP. Finally, extensive ablation studies are conducted to analyze the characteristics of MSSFP.

TABLE I
Number of Training and Testing Samples for IP

Class	Name	Training	Testing
1	Corn Notill	50	1384
2	Corn Mintill	50	784
3	Corn	50	184
4	Grass Pasture	50	447
5	Grass Trees	50	697
6	Hay Windrowed	50	439
7	Soybean Notill	50	918
8	Soybean Mintill	50	2418
9	Soybean Clean	50	564
10	Wheat	50	162
11	Woods	50	1244
12	Buildings Grass Trees Drives	50	330
13	Stone Steel Towers	50	45
14	Alfalfa	15	39
15	Grass Pasture Mowed	15	11
16	Oats	15	5
	Total	695	9671

TABLE II
Number of Training and Testing Samples for HU

Class	Name	Training	Testing
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot1	192	1041
13	Parking Lot2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
	Total	2832	12197

TABLE III
Number of Training and Testing Samples for PUS

Class	Name	Training	Testing
1	Asphalt	548	6304
2	Meadows	540	18146
3	Gravel	392	1815
4	Trees	524	2912
5	Metal Sheets	265	1113
6	Bare Soil	532	4572
7	Bitumen	375	981
8	Bricks	514	3364
9	Shadows	231	795
	Total	3921	40002

A. Datasets

1) *Indian Pines:* This HS image was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1992. It roughly covers the Purdue University Agronomy farm in Northwestern Indiana,

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TABLE IV
CLASSIFICATION PERFORMANCE OF ALL COMPARISON METHODS ON IP

Class	3-D CNN	RNN	SSTN	ViT	SF	morphFormer	GMA-Net	MAEST	SC-SS-MTr	MSSFP
1	**79.70±8.39**	69.44±5.86	55.13±23.06	54.77±2.65	51.66±12.14	74.75±11.95	65.49±27.92	50.58±7.05	70.82±22.89	67.99±3.53
2	81.12±10.66	62.24±10.67	65.43±23.67	60.33±6.24	78.95±6.54	76.43±6.23	65.71±42.08	**92.09±4.65**	69.52±27.97	87.24±3.17
3	90.22±11.75	88.04±3.51	38.59±1.86	86.96±6.46	90.22±3.54	99.13±1.42	**99.78±0.49**	98.91±5.27	99.67±0.49	91.30±7.85
4	88.81±4.27	80.54±8.80	91.05±35.24	88.37±3.80	**93.51±9.69**	76.73±15.53	93.29±1.52	88.14±3.86	83.89±11.54	92.17±2.63
5	86.23±3.13	78.62±5.08	96.84±11.57	84.65±1.37	78.91±5.82	83.67±29.76	**98.31±1.26**	89.38±3.91	96.13±7.54	90.67±3.39
6	98.86±1.59	97.72±8.13	**100.00±0.00**	88.84±3.04	95.67±2.82	99.04±1.51	99.91±0.20	96.81±4.57	99.23±1.38	99.32±1.36
7	54.03±18.28	60.68±17.19	81.37±17.54	**84.86±7.01**	74.29±3.90	68.34±17.05	43.92±40.79	82.79±2.29	50.63±44.57	82.79±3.22
8	71.92±4.94	65.80±7.91	**79.49±17.11**	67.20±7.10	75.85±12.59	71.80±9.98	73.82±32.52	76.05±5.08	78.22±14.39	79.32±7.66
9	73.94±10.68	67.20±6.22	67.55±38.13	58.87±8.79	66.31±3.90	66.24±22.28	**79.15±10.55**	70.74±5.10	70.64±19.37	71.45±4.86
10	**100.00±0.00**	99.38±2.78	**100.00±0.00**	98.77±1.01	98.15±0.28	99.88±0.28	99.88±0.28	**100.00±0.00**	**100.00±0.00**	99.38±0.28
11	62.54±4.93	93.01±12.93	87.86±5.92	85.53±5.62	94.13±5.01	95.10±4.32	**99.29±0.66**	90.68±2.35	95.98±4.37	92.60±3.54
12	**100.00±0.00**	61.52±11.25	**100.00±0.00**	60.61±4.98	82.73±9.50	91.70±7.74	99.03±1.58	90.00±2.31	96.12±3.60	73.33±3.69
13	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	97.78±1.99	97.78±2.90	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	96.89±4.87	**100.00±0.00**
14	87.18±7.82	71.79±37.64	84.62±16.18	84.62±4.21	69.23±16.66	78.97±18.70	6.15±13.76	89.74±11.41	71.79±20.59	**94.87±12.11**
15	**100.00±0.00**	**100.00±0.00**	81.82±40.96	90.91±4.07	**100.00±0.00**	94.55±8.13	20.00±44.72	**100.00±0.00**	27.27±42.15	**100.00±0.00**
16	**100.00±0.00**	80.00±8.94	**100.00±0.00**	60.00±20.00	**100.00±0.00**	80.00±44.72	20.00±44.72	**100.00±0.00**	16.00±35.78	**100.00±0.00**
OA (%)	76.04±2.07	73.25±2.96	78.53±3.28	72.55±1.73	77.22±4.04	79.12±5.20	78.12±0.42	79.90±2.55	79.54±1.30	**82.79±1.29**
AA (%)	85.91±2.31	79.75±5.13	83.11±4.28	78.32±2.45	84.21±2.20	84.77±5.75	72.73±8.71	88.49±1.58	76.43±3.15	**88.90±1.16**
Kappa (%)	72.79±2.37	69.48±3.25	75.54±3.89	68.82±1.90	73.92±4.39	76.23±5.96	74.82±0.72	77.08±2.84	76.50±1.59	**80.39±1.41**

USA. The spatial size of this image is 145 × 145 and the resolution is 20 m. The original 220 bands covering the wavelength range of 400–2500 nm are reduced to 200 bands when 20 noisy and water absorption bands are removed. The land-cover objects in the scene are grouped into 16 categories and the total number of labeled samples is 10 366.

2) *Houston:* This dataset records the scene of the University of Houston by the Innovation, Technology, Research, Excellence, and Service (ITRES) Compact Airborne Spectrographic Imager (CASI)-1500 sensor. The generated HS image contains 349 × 1905 pixels and 144 spectral bands ranging from 364 to 1046 nm. Here, the cloud is removed according to [40]. Fifteen classes of land-cover objects are included in this HS image.
3) *Pavia University:* The Reflective Optics System Imaging Spectrometer (ROSIS) sensor captures the scene over the University of Pavia and this HS image is obtained. The spatial resolution of this HS image is 1.3 m and it is composed of 610 × 340 pixels. This dataset includes 43 923 labeled samples which belong to nine categories. It contains 103 spectral bands.

B. Implementation Details

The proposed MSSFP is implemented with PyTorch on a personal computer with a CPU of i7-10700 and a GPU of NVIDIA Geforce RTX 3090. Adam [56] is used to optimize the network with a batch size of 64 for both pretraining and fine-tuning. More specifically, learning rate is set to $1e^{-3}$ and $5e^{-4}$ for pretraining and fine-tuning, respectively. Besides, the learning rate used in the pretraining stage is multiplied by 0.9 every (1/10) of the total epoch. The total epochs of pretraining and fine-tuning are separately set to 300 and 400 for three datasets. The traditional transformer is an encoder–decoder network and we use an encoder of transformer in MSSFP. All the comparison methods are reimplemented with PyTorch according to the public codes released by the authors. To assess the classification performance of all comparison methods, overall accuracy (OA), average accuracy (AA), and Kappa coefficient are used. All methods are repeated five times and the average results are reported to reduce the effect of random selection. We divide labeled samples into training and testing ones by following the setup of [4], [40], [52]. The detailed numbers of training and testing samples are listed in Tables I–III.

C. Comparison With State-of-the-Art Classification Methods

To validate the effectiveness of MSSFP, we compare it with 3-D CNN [57], RNN [17], SSTN [34], ViT [35], SF [40], morphFormer [58], GMA-Net [59], MAEST [52], and SC-SS-MTr [4]. The classification accuracies of these methods are listed in Table IV. It can be observed that the accuracy obtained by ViT is slightly lower than that by RNN, probably because ViT contains more parameters than RNN. Compared with RNN, 3-D CNN improves OA, AA, and Kappa by 2.79%, 6.16%, and 3.31%, respectively. This improvement owes to the use of an HS cube with abundant spatial information. However, 3-D CNN is still inferior to SSTN, ViT, SF, morphFormer, GMA-Net, MAEST, and SC-SS-MTr in terms of OA, AA, and Kappa because it only considers the local dependency within a fixed neighborhood. By introducing a transformer, the latter could capture long-range dependencies. Among transformer-based methods, MAEST and SC-SS-MTr achieve the best performance in terms of all metrics. As mentioned before, it is difficult to train a transformer well for its numerous parameters and limited training samples. To handle this problem, MAEST and SC-SS-MTr design a *mask-and-predict* task to pretrain the transformer without any labels. In comparison with the second-best method MAEST, MSSFP improves OA from 79.90% to 82.79%, AA from 88.49% to 88.90%, and Kappa from 77.08% to 80.39%. We conjecture that MSSFP is superior to MAEST in two aspects. On the one hand, MAEST predicts the original pixel values of masked regions, easily leading to an ambiguous

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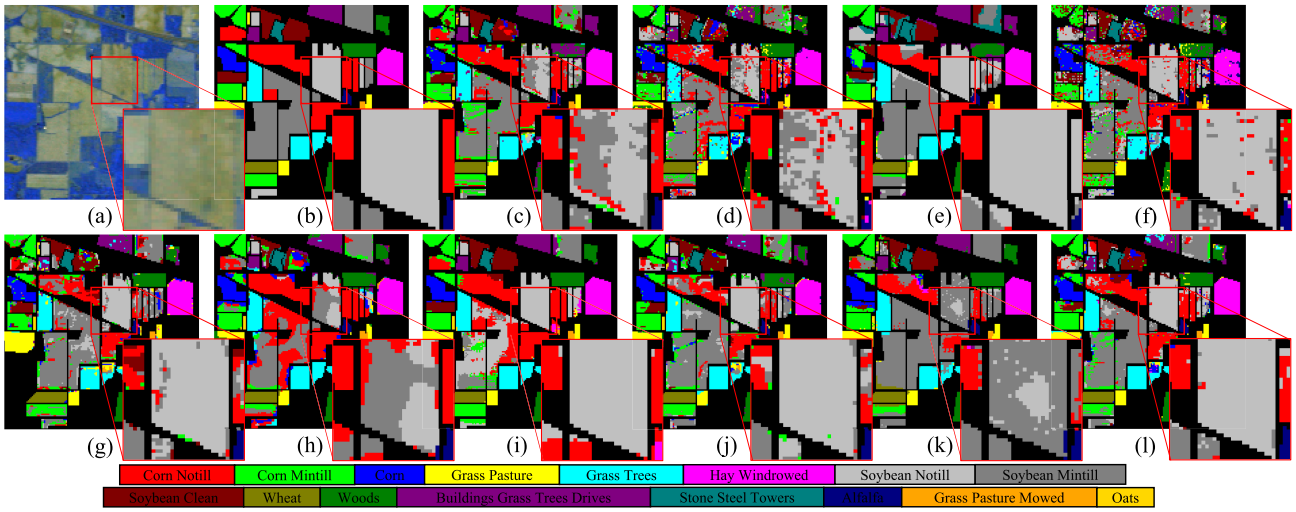

Fig. 3. Ground truth and classification maps of different methods for IP. (a) False-color map. (b) Ground truth. (c) Three-Dimensional CNN. (d) RNN. (e) SSTN. (f) ViT. (g) SF. (h) morphFormer. (i) GMA-Net. (j) MAEST. (k) SC-SS-MTr. (l) MSSFP.

TABLE V
CLASSIFICATION PERFORMANCE OF ALL COMPARISON METHODS ON HU

Class	3-D CNN	RNN	SSTN	ViT	SF	morphFormer	GMA-Net	MAEST	SC-SS-MTr	MSSFP
1	76.35±0.56	**83.48±1.57**	83.10±2.31	82.24±0.68	85.28±0.79	82.05±0.78	81.61±0.74	82.62±0.94	81.52±6.87	82.81±0.65
2	**99.81±0.39**	97.74±1.36	96.05±1.85	90.70±2.67	97.18±3.86	97.67±3.68	96.05±2.72	99.34±1.02	98.06±2.47	98.21±0.42
3	89.50±2.83	99.80±0.52	**100.00±0.00**	99.60±0.11	91.29±4.36	97.74±0.33	94.57±8.13	89.70±2.11	83.52±10.97	94.85±5.77
4	98.11±0.59	99.62±1.77	95.74±5.23	99.43±0.58	98.58±1.69	95.40±1.95	**99.64±0.19**	98.01±1.14	94.30±4.96	98.48±0.74
5	97.06±0.29	91.29±2.30	97.73±7.83	97.73±0.05	**99.81±0.72**	98.45±0.43	99.60±0.74	99.24±0.59	96.97±5.64	99.62±0.53
6	75.52±2.37	95.10±0.31	**95.80±0.77**	95.10±0.00	95.10±4.33	91.47±8.48	87.97±9.24	93.71±3.28	90.07±5.73	**95.80±0.91**
7	**95.62±2.08**	70.15±4.94	52.71±6.63	83.77±1.85	88.71±4.25	94.12±3.33	91.60±2.50	88.53±2.18	80.97±4.68	87.78±0.55
8	69.90±1.89	55.08±7.46	82.53±10.46	55.18±3.79	78.16±10.21	72.52±10.67	58.73±18.88	83.10±8.33	**87.29±5.39**	83.19±9.12
9	72.24±4.26	74.41±5.91	**86.78±14.12**	66.19±6.63	77.43±5.92	76.21±3.05	76.83±8.84	73.94±3.26	79.47±12.60	72.90±4.66
10	52.51±12.59	63.32±17.74	84.17±14.36	63.90±6.34	60.42±4.63	57.22±21.89	79.50±14.31	73.36±13.11	85.50±9.93	**87.64±7.94**
11	85.10±5.66	90.80±3.97	75.62±8.41	54.74±7.97	83.02±2.68	81.48±10.13	**93.55±12.30**	76.85±3.16	91.25±15.07	82.73±3.74
12	**83.67±6.10**	63.02±16.98	72.91±17.64	49.66±3.73	72.05±3.89	71.13±8.36	65.44±23.05	80.02±2.76	79.63±9.98	82.90±4.52
13	71.93±1.20	44.21±9.04	38.60±12.99	61.40±4.11	73.68±7.34	81.96±4.14	82.81±7.08	64.56±8.60	**83.72±5.50**	81.05±5.52
14	96.36±2.23	98.79±0.34	98.79±0.68	99.60±0.61	99.19±1.82	90.69±7.19	**100.00±0.00**	96.36±2.60	**100.00±0.00**	97.17±0.79
15	67.44±6.24	98.10±1.38	98.10±0.91	98.52±0.24	**99.58±2.13**	94.25±6.76	97.42±2.96	95.98±8.23	94.50±5.66	98.31±1.02
OA (%)	82.70±0.97	80.33±1.96	83.47±1.29	76.86±0.95	85.23±2.08	84.03±1.65	85.57±1.27	85.95±1.73	87.81±0.86	**88.49±0.69**
AA (%)	82.07±1.00	81.66±1.36	83.91±1.33	79.85±0.85	86.63±2.23	85.49±1.58	87.02±1.52	86.35±1.39	88.45±0.93	**89.56±0.47**
Kappa (%)	81.20±1.05	78.67±2.10	82.05±1.41	74.96±1.03	83.96±2.25	82.67±1.79	84.34±1.38	84.73±1.88	86.77±0.92	**87.50±0.75**

TABLE VI
CLASSIFICATION PERFORMANCE OF ALL COMPARISON METHODS ON PUS

Class	3-D CNN	RNN	SSTN	ViT	SF	morphFormer	GMA-Net	MAEST	SC-SS-MTr	MSSFP
1	93.96±2.75	80.79±5.07	**99.65±9.51**	78.70±2.68	83.19±4.49	85.87±5.42	82.65±7.49	79.28±3.66	87.27±6.98	87.42±1.51
2	**98.05±0.69**	77.24±7.63	87.53±5.63	62.61±4.96	69.86±5.71	93.22±5.92	97.55±3.47	93.65±6.01	92.59±7.46	94.94±1.08
3	69.31±3.24	54.55±17.14	63.14±39.15	64.90±2.54	75.10±19.50	74.09±15.11	48.54±29.33	73.33±4.56	**99.32±0.83**	83.75±3.91
4	97.15±0.42	**98.49±2.84**	94.71±2.60	95.95±1.49	92.24±1.24	93.74±1.69	96.56±1.74	97.94±0.17	96.33±1.60	95.43±0.60
5	97.66±0.62	99.46±0.30	99.82±0.61	99.37±0.19	98.65±0.21	97.77±2.49	97.32±5.94	**99.91±0.49**	95.71±6.05	99.10±0.16
6	28.65±5.31	53.24±16.76	71.15±16.41	90.53±9.70	**98.32±5.15**	80.64±12.68	61.22±7.85	86.07±10.04	88.04±16.07	76.47±7.28
7	52.60±4.93	90.42±3.31	39.65±1.91	87.56±1.77	91.03±5.98	74.15±27.30	89.66±7.74	90.21±6.85	89.13±21.35	**91.34±1.55**
8	79.46±1.88	91.32±5.84	85.61±21.14	84.48±5.23	92.00±7.58	87.56±8.39	**99.23±0.44**	97.06±2.09	71.84±28.45	95.12±0.79
9	92.20±2.15	**99.62±0.62**	93.21±1.10	98.36±0.17	92.08±2.69	96.23±2.10	96.58±7.44	93.58±0.98	97.38±1.50	99.50±1.62
OA (%)	85.30±0.58	78.14±1.46	86.11±2.33	75.05±1.90	80.70±3.18	89.04±4.27	88.68±0.87	90.29±3.09	90.16±0.70	**91.31±0.95**
AA (%)	78.78±0.88	82.79±2.41	81.61±3.78	84.72±1.24	88.05±3.65	87.03±4.76	85.48±3.92	90.11±1.66	90.85±4.41	**91.45±1.25**
Kappa (%)	79.41±0.83	71.32±1.48	81.39±3.05	68.78±2.04	75.53±3.94	85.32±5.57	84.48±1.18	87.01±3.94	86.87±0.76	**88.26±1.34**

problem. To alleviate this problem, MSSFP chooses to predict the hand-crafted features of masked regions for better capturing of global statistics. On the other hand, MSSFP uses an adaptive aggregation strategy to adaptively fuse two results obtained by input spectral and spatial sequences. But MAEST extracts features from one input sequence which mixes together spectral and spatial information, ignoring the correlation between them. Classification maps of all methods

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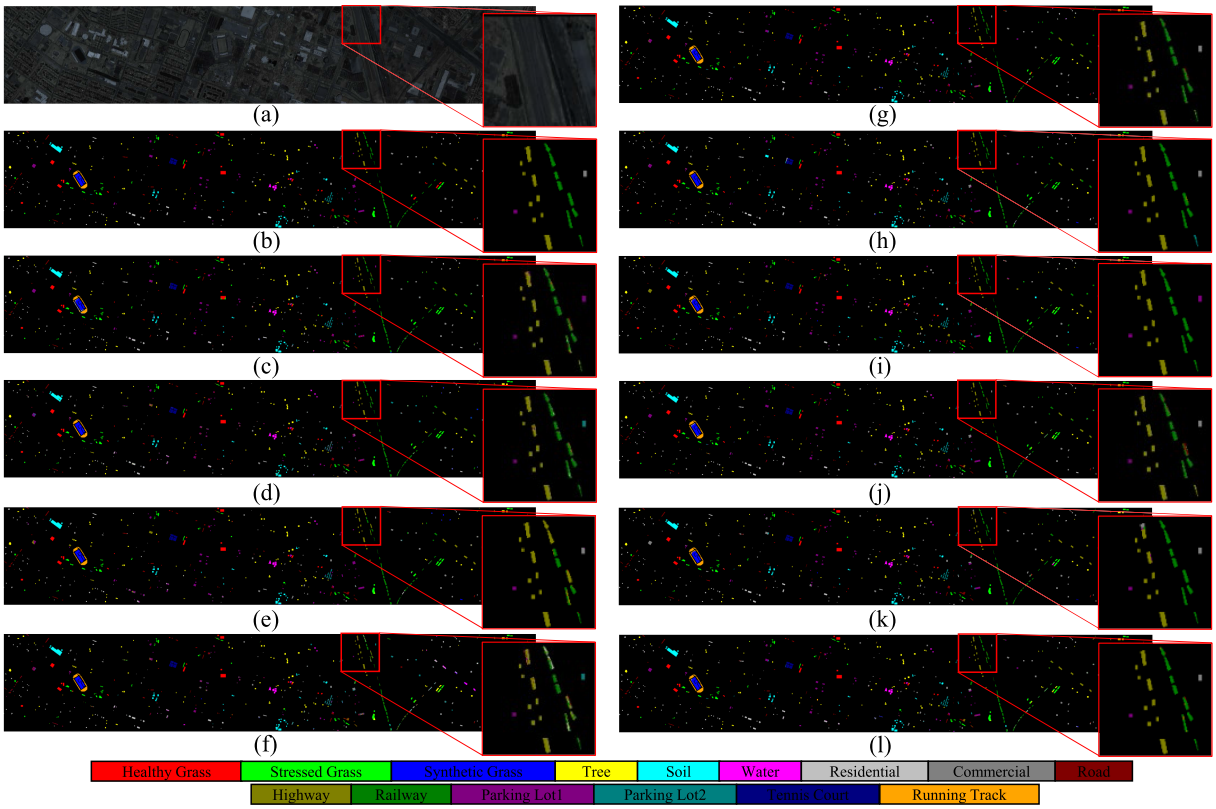

Fig. 4. Ground truth and classification maps of different methods for HU. (a) False-color map. (b) Ground truth. (c) Three-Dimensional CNN. (d) RNN. (e) SSTN. (f) ViT. (g) SF. (h) morphFormer. (i) GMA-Net. (j) MAEST. (k) SC-SS-MTr. (l) MSSFP.

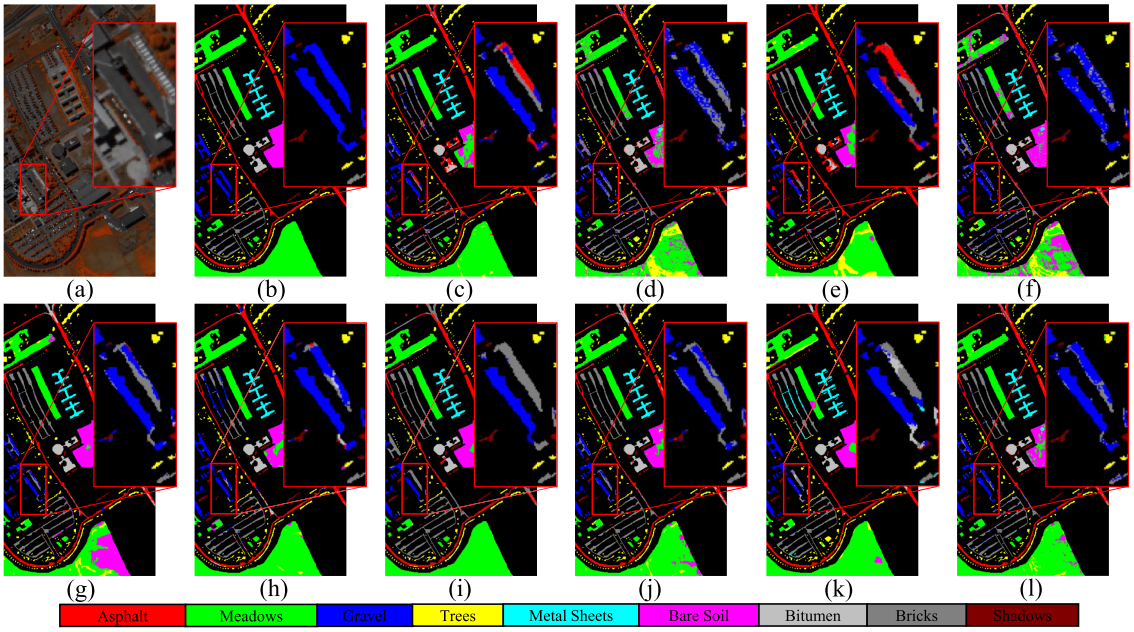

Fig. 5. Ground truth and classification maps of different methods for PUS. (a) False-color map. (b) Ground truth. (c) Three-Dimensional CNN. (d) RNN. (e) SSTN. (f) ViT. (g) SF. (h) morphFormer. (i) GMA-Net. (j) MAEST. (k) SC-SS-MTr. (l) MSSFP.

are shown in Fig. 3. Taking the marked region as an example, all comparison methods fail to distinguish *corn notill*, *soybean notill*, and *soybean mintill* while our proposed MSSFP does.

For HU and PUS, we can draw the similar conclusion from Tables V and VI. MAEST and SC-SS-MTr obtain better classification performance than other comparison methods. In comparison with the second-best method SC-SS-MTr, MSSFP further improves OA, AA, and Kappa by 0.68%, 1.11%, and 0.73% on HU. For PUS, MSSFP further improves OA, AA, and Kappa of the second-best method MAEST by 1.02%, 1.34%, and 1.25%. Classification maps on these two datasets are shown in Figs. 4 and 5. Similarly, MSSFP is

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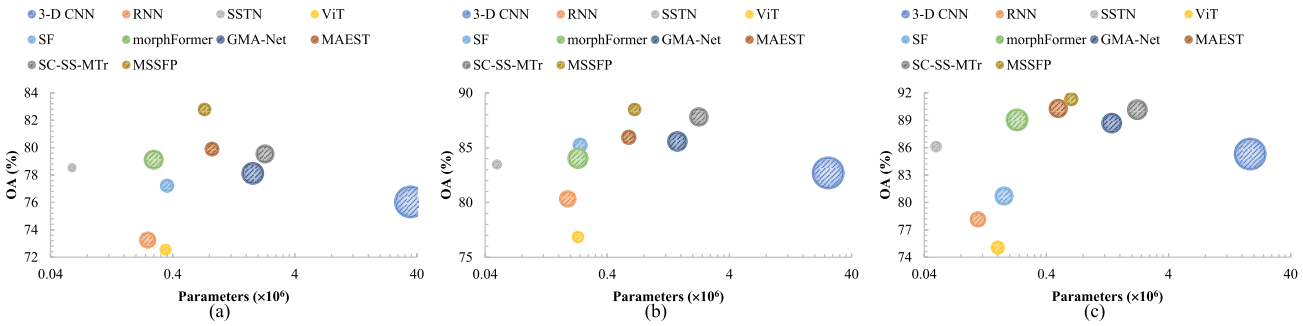

Fig. 6. Computational complexity (network parameters and inference time for each testing sample) of different methods on three datasets. (a) IP. (b) HU. (c) PUS. Note that the inference time is illustrated by the radius of the circle.

able to distinguish land-cover objects of different categories, generating a better satisfactory result.

In addition, we compare MSSFP with other methods in terms of computational complexity. The result is shown in Fig. 6. It can be observed that MSSFP outperforms *mask-and-predict* methods (MAEST and SC-SS-MTr) with respect to both classification performance and computational complexity on IP. Although parameters of some methods like morphFormer are less than that of MSSFP, the improvement ($\geq$3.67%) of OA compensates for that. Similarly, the performance of comparison methods that contains more parameters are worse than that of MSSFP on HU and PUS. Compared with the second-best method MAEST, MSSFP contains a few more parameters but it remarkably improves OA by 1.02% and 2.54% on HU and PUS. In a word, MSSFP achieves a good balance between classification performance and computational complexity.

D. Ablation Study

1) Effectiveness of Each Component: In MSSFP, two key components are proposed to boost the performance of the transformer on HS image classification. Next, we conduct several experiments on the above three datasets to validate the effectiveness of two components. Table VII lists the experimental results of MSSFP with different configurations on three datasets. Herein, we denote a pure spectral–spatial transformer without any extra components as *baseline*. We can observe that *baseline* obtains the lowest OA on IP, HU, and PUS without adding any components. Once the *mask-and-predict* task is integrated into *baseline*, OAs on three datasets are separately improved by 3.05%, 1.91%, and 1.32%. When the proposed adaptive aggregation strategy is used, OAs on three datasets are improved by 2.27%, 1.51%, and 0.53%, respectively. It can be observed that the improvement brought by the adaptive aggregation strategy is less than that by the *mask-and-predict* task. This phenomenon indicates that the classification performance of MSSFP heavily depends on the *mask-and-predict* task. If two components are simultaneously used, MSSFP achieves the highest classification performance. Furthermore, we test the performance of MSSFP with a pixel-wise *mask-and-predict* task. As shown in Table VII, pixel prediction decreases OA obtained by feature prediction by 0.64%, 0.51%, and 0.31% on IP, HU, and PUS, respectively.

TABLE VII
PERFORMANCE OF MSSFP WITH DIFFERENT CONFIGURATIONS ON THREE DATASETS

Model configuration	aa[1]		✓		✓	✓
	map[2](pixel)				✓	
	map[2](feature)			✓		✓
Dataset	IP	78.41	80.68	81.46	82.15	**82.79**
	HU	85.24	86.75	87.15	87.98	**88.49**
	PUS	89.64	90.17	90.96	91.00	**91.31**

[1] adaptive aggregation
[2] *mask-and-predict*

TABLE VIII
EFFECT OF DIFFERENT PATCH SIZES ON THREE DATASETS

Dataset	1×1	3×3	5×5	7×7	9×9	11×11
IP	79.84	80.48	81.07	**82.79**	82.11	80.63
HU	86.07	86.97	87.86	**88.49**	87.60	86.37
PUS	90.48	**91.31**	90.73	90.01	89.17	88.22

Herein, we illustrate the predicted features of spectral and spatial domains on three datasets. Three rows in Fig. 7 correspond to the MSSFP of IP, HU, and PUS, respectively. The first column is the masked spectral feature prediction and the second, third, and fourth columns are masked spatial feature predictions. To better illustrate the predicted principal components of the spectrum, we flatten the spatial patches of three principal components to three vectors. Then, they are concatenated together and plotted as a line chart. As shown in the figure, MSSFP accurately predicts the first three principal components of all spectral bands even though 75% of input spectral bands are masked. Meanwhile, we randomly select three masked HS pixels to show the quality of predicted LBP features. It is shown that the trend of the line chart predicted by MSSFP is almost consistent with that of ground truth. It is noteworthy that although MSSFP is unable to predict masked spatial features as accurately as spectral features, this *mask-and-predict* task has already helped MSSFP achieve a competitive classification result, in comparison with state-of-the-art methods.

2) Hyperparameters Selection: There exist three main hyperparameters in MSSFP, including patch size, masking ratio, and σ. First, we fix other hyperparameters and select the optimal patch size from a candidate set $\{1 \times 1, 3 \times 3,$

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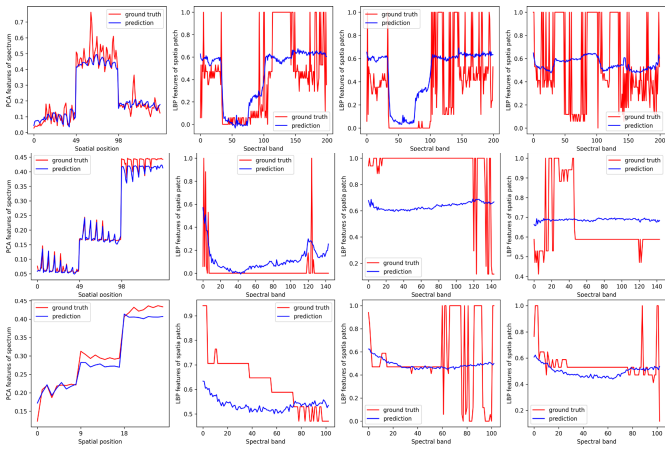

Fig. 7. Illustration of the MSSFP on (first row) IP, (second row) HU, and (third row) PUS.

TABLE IX
CLASSIFICATION PERFORMANCE WITH DIFFERENT MASKING RATIOS

Dataset	0%	15%	35%	55%	75%	95%
IP	80.82	81.78	81.98	82.43	**82.79**	82.31
HU	85.47	86.27	87.62	88.07	**88.49**	88.13
PUS	89.99	90.25	90.72	91.11	**91.31**	90.96

TABLE X
EFFECT OF DIFFERENT σ VALUES ON THREE DATASETS

Dataset	0.0005	0.005	0.05	0.5	5
IP	80.07	81.18	82.24	**82.79**	82.68
HU	85.45	86.83	87.50	**88.49**	88.43
PUS	88.62	89.77	90.48	**91.31**	91.13

$5 \times 5, 7 \times 7, 9 \times 9, 11 \times 11\}$. The experimental result is listed in Table VIII. For IP and HU, OA achieves the highest value when the patch size is set to 7×7. For PUS, the optimal patch size is set to 3×3 and MSSFP obtains the best OA. Then, the masking ratio is chosen from six possible values $\{0\%, 15\%, 35\%, 55\%, 75\%, 95\%\}$. It can be observed in Table IX that OA will reach the highest value when the masking ratio is set to 75%. This masking ratio is consistent with existing *mask-and-predict*-based methods like MAE [51] and MAEST [52]. Moreover, MSSFP could obtain a similar OA to MAEST even though MSSFP is pretrained without masking. This demonstrates the effectiveness of hand-crafted feature prediction and adaptive aggregation strategy. Finally, we test the effect of different σ values on three datasets. As shown in Table X, OA will improve and then decrease as the σ value increases. When the σ value is set to 0.5, MSSFP produces the best classification results.

Furthermore, we conduct experiments to determine the optimal hyperparameters of the transformer. The key hyperparameters include the number of heads, feature dimension, and the number of MSA. Specifically, they are selected from three candidate sets $\{1, 2, 4, 8, 16, 32\}$, $\{8, 16, 32, 64, 128, 256\}$, and $\{1, 2, 3, 4, 5, 6\}$, respectively. The experimental results of different hyperparameters on three datasets are shown in Tables XI–XIII. When the number of heads, feature dimension, and the number of MSA are separately set to 4, 64, and 5, the OA of MSSFP reaches the highest on all three datasets.

TABLE XI
EFFECT OF DIFFERENT NUMBERS OF HEADS IN TRANSFORMER ON THREE DATASETS

Dataset	1	2	4	8	16	32
IP	79.85	81.60	**82.79**	82.50	80.98	78.69
HU	86.96	87.13	**88.49**	87.79	86.62	86.40
PUS	90.49	91.05	**91.31**	91.18	90.08	89.79

TABLE XII
EFFECT OF DIFFERENT FEATURE DIMENSIONS IN TRANSFORMER ON THREE DATASETS

Dataset	8	16	32	64	128	256
IP	81.74	82.01	82.43	**82.79**	82.62	82.03
HU	87.51	87.77	87.95	**88.49**	88.12	87.92
PUS	90.67	90.88	91.08	**91.31**	90.99	90.85

TABLE XIII
EFFECT OF DIFFERENT NUMBERS OF MSA IN TRANSFORMER ON THREE DATASETS

Dataset	1	2	3	4	5	6
IP	80.47	81.24	81.72	82.14	**82.79**	82.17
HU	86.07	86.81	87.39	87.83	**88.49**	88.06
PUS	88.71	89.63	90.27	90.88	**91.31**	90.85

V. CONCLUSION

In this article, we propose a novel HS image classification method called MSSFP. MSSFP first pretrains a transformer-based encoder–decoder network by predicting the hand-crafted features of masked regions in spectral and spatial dimensions without labels. Then, the encoder of the network is adapted for HS image classification by fine-tuning it with limited training samples. Besides, the correlation between spectral and spatial domains is captured by a proposed strategy of adaptive aggregation. Due to the *mask-and-predict* task and adaptive aggregation strategy, we can give full play to the transformer in HS image classification. Experimental results on three public datasets demonstrate the effectiveness of MSSFP.

Even though feature prediction is superior to pixel prediction for *mask-and-predict* task, the hand-crafted features used in MSSFP might contain inadequate discriminant information. Besides, the effectiveness of *mask-and-predict* task is influenced not only by predicted targets but also by other factors like task difficulty. In the future, we will study the effect of different kinds of features and other possible factors on *mask-and-predict* task, and modify this task to be more suitable for learning spectral–spatial features from HS images.

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